HUMAN CARDIOVASCULAR PHYSIOLOGY • MEDICAL INFOGRAPHIC

THE HUMAN HEART: ARCHITECTURE, HEMODYNAMICS & SYSTEMIC ORGAN PERFUSION

A Complete Illustrated Infographic Blueprint: Anatomical Chambers, Valvular Mechanics, Electrical Vectors, Systemic Organ Delivery & Clinical Biomarkers

MECHANICAL PUMP
7,500 L / Day Fluid Displacement

Cardiac Output: 4.5 - 5.5 L/min at rest; surges to 20-25 L/min during peak physical exertion.

Stroke Volume: ~70 mL blood ejected per beat into the ascending aorta and pulmonary trunk.

Lifetime Workload: Beats ~100,000 times/day and over 2.5 billion times in an average lifespan.

Hemodynamic Role: Maintains the systemic arterial pressure gradient (MAP >65 mmHg) powering microvascular filtration.

ELECTRICAL OSCILLATOR
Autonomous Pacemaker Grid

SA Node Pacemaker: Spontaneous phase 4 diastolic depolarization at 60-100 actions potentials/min.

AV Nodal Gate: Imposes a critical 0.12-second delay allowing complete ventricular end-diastolic filling.

Fast Conduction: Purkinje fiber network deploys electrical vector at 2-4 m/s apex-to-base.

Syncytium: Gap junctions in intercalated discs allow simultaneous all-or-nothing syncytial contraction.

ORGAN PERFUSION ENGINE
Lifeline to All Vital Organs

Brain Delivery: 15% of Cardiac Output (~750 mL/min); neurons suffer necrosis after 4-5 min arrest.

Renal Filtration: 20-25% of CO (~1,200 mL/min) powering GFR and electrolyte homeostasis.

Hepato-Splanchnic: 25% of CO nourishing liver detox, digestion, and nutrient absorption.

Metabolic Delivery: Delivers 1,000 mL O2/min and cleanses cellular metabolic byproducts.

ENDOCRINE GLAND
Natriuretic Peptide Regulation

ANP Hormone: Secreted by atrial myocytes in response to atrial distension and hypervolemia.

BNP Hormone: Secreted by ventricular myocardium under wall tension; diagnostic gold standard for heart failure.

Target Action: Promotes renal natriuresis, diuresis, systemic vasodilation, and RAAS suppression.

Fluid Homeostasis: Acts as the intrinsic physiological brake preventing lethal circulatory overload.

Authoritative Source Standards: Guyton & Hall Medical Physiology (14th Ed.) | Braunwald's Heart Disease | Harrison's Principles of Internal Medicine

Four-Chamber Architecture, Myocardial Walls & Pressure Gradients

Asymmetrical muscular architecture matches differing hemodynamic workloads: LV generates 5x RV peak systolic pressure.

RIGHT ATRIUM (RA)

Low-Pressure Venous Collector

Inflow Sources: Receives deoxygenated systemic venous return from Superior Vena Cava (head/arms), Inferior Vena Cava (body/legs), and Coronary Sinus (myocardium).

Normal Pressure: Central Venous Pressure (CVP) ranges 2 - 6 mmHg.

Internal Structures: Fossa ovalis (remnant of fetal foramen ovale), crista terminalis, and pectinate muscles lining the right auricle.

Electrophysiology: Houses Sinoatrial (SA) node in high posterior wall near SVC junction and Atrioventricular (AV) node near triangle of Koch.

RIGHT VENTRICLE (RV)

Low-Resistance Pulmonary Pump

Geometry: Crescent-shaped cavity wrapped around the muscular interventricular septum.

Pressure Profile: Systolic 15 - 30 mmHg; End-diastolic 2 - 8 mmHg.

Outflow Conduit: Pumps blood through Pulmonary Semilunar Valve into the low-resistance pulmonary trunk toward alveolar capillary beds.

Wall Thickness: Thin muscular wall (3 - 5 mm) calibrated for high capacitance and low impedance of the pulmonary vascular circuit.

LEFT ATRIUM (LA)

High-Oxygen Inflow Reservoir

Inflow Sources: Receives fully oxygenated blood (98% SaO2) from the lungs via 4 Pulmonary Veins (2 superior, 2 inferior; valve-less).

Normal Pressure: Mean pressure 4 - 12 mmHg (surrogate for Left Ventricular End-Diastolic Pressure).

Endocrine Secretion: Atrial distension stimulates release of Atrial Natriuretic Peptide (ANP).

Clinical Significance: Left atrial enlargement predisposes to Atrial Fibrillation and thromboembolic stroke formation in left atrial appendage.

LEFT VENTRICLE (LV)

High-Pressure Systemic Powerhouse

Geometry & Wall: Bullet-shaped conical chamber with thick myocardium (10 - 15 mm, 3x thicker than RV).

Pressure Profile: Peak systolic 100 - 140 mmHg; End-diastolic 3 - 12 mmHg.

Outflow Conduit: Overcomes Systemic Vascular Resistance (SVR) to propel blood through Aortic Valve into the Ascending Aorta.

Wall Layers: Endocardium (smooth inner endothelium), Myocardium (helical spiraling muscle fibers producing wringing systolic ejection), Epicardium.

Anatomical References: Netter's Atlas of Human Anatomy (8th Ed.) | Guyton & Hall Medical Physiology, Ch. 9.

Heart Valve Mechanics: Unidirectional Flow, Fibrous Skeleton & Auscultation

Passive pressure-operated flap valves guarantee strictly unidirectional blood flow; closure events produce the classic Lub-Dub (S1/S2) heart sounds.

TRICUSPID VALVE

Right Atrioventricular (AV) Gate

Anatomy: Three fibrous cusps (Anterior, Posterior, Septal) spanning ~7-9 cm² orifice.

Tethering System: Chordae tendineae connect cusps to papillary muscles in the RV wall.

Closing Mechanics: During ventricular systole, rising RV pressure forces cusps together; papillary muscle contraction tensions chordae to prevent prolapse into RA.

Auscultation: Component of S1 heart sound; best heard at the lower left sternal border (4th-5th intercostal space).

PULMONARY VALVE

Right Ventricular Outflow Gate

Structure: Three pocket-shaped semilunar cusps (Anterior, Left, Right) with central nodules of Arantius.

Operation: Opens when RV pressure exceeds pulmonary trunk diastolic pressure (>10-15 mmHg).

Diastolic Seal: Snaps shut when RV relaxes and pulmonary arterial column recoils backward, filling the pocket cusps.

Auscultation: Closure forms P2 component of S2 sound; physiological inspiratory splitting occurs due to increased right-sided venous return.

MITRAL / BICUSPID VALVE

Left Atrioventricular (AV) Gate

Anatomy: Two large, durable cusps (Anterior and Posterior) anchored to prominent anterolateral and posteromedial papillary muscles.

Peak Workload: Withstands peak left ventricular systolic pressure (120-140 mmHg) without allowing regurgitation.

Closure Dynamics: Snaps shut at the very onset of ventricular isovolumetric contraction; produces the loud initial M1 component of S1 ('LUB').

Clinical Trap: Mitral valve prolapse (MVP) and rheumatic mitral stenosis represent the most frequent valvular pathologies worldwide.

AORTIC VALVE

High-Pressure Systemic Gateway

Structure: Three thick semilunar cusps (Right, Left, Posterior/Non-coronary) seated within the aortic root.

Sinuses of Valsalva: Anatomical pouches behind the right and left cusps house the ostia of the Right and Left Main Coronary Arteries.

Diastolic Closure: Closure produces the crisp A2 component of S2 ('DUB'); diastolic recoil directly perfuses the coronary vascular bed!

Hemodynamic Metric: Normal orifice area is 3.0 - 4.0 cm²; severe aortic stenosis (<1.0 cm²) drastically elevates LV afterload.

Dual Circulation: Pulmonary Circuit vs. Systemic High-Pressure Vascular Circuit

William Harvey's closed dual circuit: Right heart drives gas exchange at 1/6th systemic pressure; Left heart propels life-sustaining perfusion to peripheral beds.

PULMONARY CIRCUIT (LOW-PRESSURE)

Alveolar Gas Exchange Loop

Vascular Path: Right Ventricle -> Pulmonary Trunk -> Left & Right Pulmonary Arteries -> Lobar/Segmental Arteries -> Alveolar Capillaries -> 4 Pulmonary Veins -> Left Atrium.

Hemodynamic Profile: Low pressure, low resistance (Mean PAP ~14 mmHg; Systolic/Diastolic 25/10 mmHg). Pulmonary vascular resistance (PVR) is 1/10th of SVR.

Gas Exchange: Deoxygenated blood (pO2 ~40 mmHg, pCO2 ~46 mmHg) offloads CO2 across the 0.5-µm alveolar-capillary membrane and binds O2 to reach pO2 ~100 mmHg (98% SaO2).

Unique Physiology: Hypoxic Pulmonary Vasoconstriction (HPV): Pulmonary vessels constrict in unventilated lung zones to shunt blood to ventilated alveoli, maximizing V/Q matching.

SYSTEMIC CIRCUIT (HIGH-PRESSURE)

Whole-Body Peripheral Perfusion Loop

Vascular Path: Left Ventricle -> Aortic Root -> Ascending/Arch/Descending Aorta -> Elastic Arteries -> Muscular Conduit Arteries -> Arterioles -> Capillaries -> Venules -> Vena Cavae.

Hemodynamic Profile: High pressure, high resistance (Systolic 120 mmHg, Diastolic 80 mmHg, Mean Arterial Pressure MAP ~93 mmHg).

Stopcock Regulation: Arterioles (rich in smooth muscle) serve as primary resistance vessels, dynamically adjusting vascular tone via autonomic and local metabolic factors.

Nutrient Delivery: Unloads O2, glucose, fatty acids, amino acids, electrolytes, and hormones; sweeps away lactic acid, urea, creatinine, and cellular metabolites.

MICROCIRCULATION & STARLING FORCES

Capillary Exchange Dynamics

Capillary Bed Surface: Over 10 billion capillaries providing a massive ~500-700 m² surface area for diffusion.

Starling Equation: Net Fluid Movement $J_v = K_f [(P_c - P_i) - \sigma (\pi_c - \pi_i)]$. Hydrostatic pressure ($P_c$) drives filtration; oncotic pressure (π_c) drives reabsorption.

Arterial Capillary End: $P_c _x0007_pprox 32 ext{ mmHg} > \pi_c _x0007_pprox 25 ext{ mmHg}$; net filtration pushes fluid, electrolytes, and nutrients into interstitial space.

Venous Capillary End: $P_c _x0007_pprox 15 ext{ mmHg} < \pi_c _x0007_pprox 25 ext{ mmHg}$; net absorption draws 85-90% of fluid back; remaining 10-15% cleared via lymphatic vessels back to venous angle.

The Cardiac Conduction System & Surface ECG Vector Mapping

Rhythmic cardiac pumping relies on an intrinsic autonomous conduction hierarchy; defects lead to lethal arrhythmias or heart block.

1. SA NODE

Primary Pacemaker

Location: Subepicardial at SVC-RA junction.

Intrinsic Rate: 60 - 100 impulses/min.

Phase 4 Drift: Funny channels (I_f) carry slow inward Na+ influx; T-type Ca2+ channels trigger threshold at -40 mV.

Vagal Dominance: Resting vagal tone suppresses intrinsic 100 bpm to resting 60-80 bpm.

2. ATRIAL TRACTS

Interatrial Wavefront

Conduction Paths: Anterior, Middle (Wenckebach), and Posterior (Thorel) internodal pathways.

Bachmann Bundle: Carries wave from RA across septum to LA ensuring synchronized bi-atrial contraction.

ECG Correlation: Corresponds to the P WAVE (atrial depolarization; normal duration <0.12s, amplitude <2.5 mm).

3. AV NODE

Physiological Gatekeeper

Location: Triangle of Koch in interatrial septum.

Critical Delay: Imposes a 0.12s delay due to slow Ca2+ conduction and fewer gap junctions.

Function: Allows complete ventricular filling prior to systole and protects ventricles from rapid atrial flutter/fibrillation.

ECG Correlation: Maps to the isoelectric PR INTERVAL (normal 0.12 - 0.20s).

4. BUNDLE OF HIS

Septal Distributor

Anatomy: Penetrates the central fibrous skeleton of the heart into muscular interventricular septum.

Bifurcation: Splits into Right Bundle Branch (RBB) and Left Bundle Branch (LBB; subdivides into Anterior & Posterior fascicles).

Speed: High conduction velocity (~2 m/s).

5. PURKINJE SYSTEM

Apex-to-Base Ejection

Network: Large specialized subendocardial myocytes with abundant gap junctions.

Velocity: Fastest conduction in heart: 2 - 4 m/s.

Apex-to-Base: Triggers ventricular depolarization from cardiac apex toward outflow base.

ECG Correlation: QRS COMPLEX (<0.12s) = ventricular depolarization; ST segment & T WAVE = ventricular repolarization.

The Cardiac Cycle: Wiggers Diagram & Pressure-Volume Relationships

Wiggers diagram integrates ECG, ventricular pressure, aortic pressure, ventricular volume, and phonocardiogram into 4 phases.

PHASE 1: ISOVOLUMETRIC CONTRACTION

Tension Rise with Constant Volume

Valvular State: ALL FOUR VALVES CLOSED (Mitral closes as LV pressure exceeds LA; Aortic valve remains closed).

Timeframe: Begins immediately following QRS complex (~0.05 seconds duration).

Pressure Jump: LV pressure soars from ~8 mmHg to 80 mmHg with zero change in chamber volume (EDV ~120 mL).

Phonocardiogram: Mitral and Tricuspid closure vibrations generate First Heart Sound (S1 'LUB').

PHASE 2: VENTRICULAR EJECTION

Rapid & Reduced Systolic Outflow

Valve Action: Aortic valve opens once LV pressure eclipses aortic diastolic pressure (>80 mmHg).

Rapid Ejection: Initial 1/3 of ejection propels 70% of stroke volume; LV pressure peaks at 120 mmHg.

Reduced Ejection: Later 2/3 shows decelerating outflow; aortic pressure declines as runoff exceeds entry.

Volume Shift: Chamber volume plummets from EDV (120 mL) to End-Systolic Volume (ESV ~50 mL). Stroke Volume ejected = 70 mL.

PHASE 3: ISOVOLUMETRIC RELAXATION

Pressure Collapse with Constant Volume

Valve Action: ALL FOUR VALVES CLOSED (Aortic valve snaps shut as LV pressure drops below aortic pressure).

Incisura / Dicrotic Notch: Brief pressure rebound in aorta caused by elastic recoil of aortic root cusps.

Pressure Plunge: LV pressure falls precipitously from ~100 mmHg down to near 0-5 mmHg.

Phonocardiogram: Aortic and Pulmonic valve closure generates Second Heart Sound (S2 'DUB').

PHASE 4: VENTRICULAR FILLING

Rapid Passive Inflow & Atrial Kick

Valve Action: Mitral valve opens when LV pressure drops below Left Atrial pressure (<8-10 mmHg).

Rapid Filling (80%): Blood stored in LA rushes passively into relaxing ventricle; suction effect of myocyte elastic recoil.

Atrial Kick (20%): Atrial systole (P wave) delivers final 15-20% boost to ventricular end-diastolic volume (EDV ~120 mL).

Clinical Trap: Loss of atrial kick in Atrial Fibrillation can cause acute decompensation in stiff, hypertrophied ventricles.

Systemic Organ Perfusion: How the Heart Serves Brain, Kidneys, Liver & Muscle

Cardiac output distribution matches metabolic demand and organ hierarchy: Brain demands continuous perfusion; Kidneys demand massive flow for filtration.

THE BRAIN

15% Cardiac Output (~750 mL/min)

Conduit Vessels: Internal Carotid & Vertebral arteries form the anastomotic Circle of Willis at brain base.

Metabolic Demand: Consumes 20% of total resting body oxygen despite accounting for only 2% of body mass!

Cerebral Autoregulation: Maintains constant blood flow across MAP range 60 - 160 mmHg via myogenic tone and local CO2/H+ vasoactivity.

Ischemic Vulnerability: Zero anaerobic capacity: syncope occurs within 6-8s of circulatory arrest; irreversible neuronal death occurs in 4-5 minutes.

THE KIDNEYS

20 - 25% Cardiac Output (~1,200 mL/min)

Massive Flow Ratio: Receives the highest blood flow per gram of tissue of any organ in the human body (~400 mL/min/100g).

Primary Purpose: Not for oxygen delivery, but to power Glomerular Filtration Rate (GFR ~125 mL/min = 180 L plasma filtered daily!).

Renal Autoregulation: Myogenic response and Tubuloglomerular Feedback (TGF via Macula Densa) preserve GFR at MAP 80-180 mmHg.

Endocrine Feedback: Hypoperfusion triggers Renin release from juxtaglomerular cells -> Angiotensin II -> Aldosterone (RAAS cascade).

LIVER & GI TRACT

25% Cardiac Output (~1,300 mL/min)

Dual Blood Supply: Hepatic Artery (25% oxygenated flow from celiac trunk) + Portal Vein (75% venous inflow draining stomach, intestines, spleen).

Functional Use: Carries absorbed carbohydrates, amino acids, and lipids directly to hepatocytes for metabolic processing, storage, and first-pass clearance.

Blood Reservoir: Hepatic sinusoids and splanchnic venules hold ~500-1,000 mL blood; sympathetic vasoconstriction mobilizes this pool during hemorrhage.

Bile & Protein Synthesis: Drives albumin synthesis (maintains plasma oncotic pressure) and coagulation factor production (I, II, VII, IX, X).

SKELETAL MUSCLE & SKIN

20% at Rest -> 85% in Heavy Exercise

Dynamic Hyperaemia: Resting flow (~1,000 mL/min) surges up to 15,000 - 20,000 mL/min during maximal exercise!

Local Vasodilation: Muscle contractions release adenosine, lactate, K+, nitric oxide (NO), and H+, dilating precapillary sphincters.

Cardiac Compensation: Heart rate rises from 70 to 180 bpm; stroke volume rises from 70 to 110 mL; CO expands from 5 to 22 L/min.

Cutaneous Shunt: Skin blood flow adjusts dynamically: dilates for radiative heat dissipation, constricts in cold to conserve core temperature.

Clinical Physiology: Boron, W. F. & Boulpaep, E. L., Medical Physiology, 3rd Ed., Elsevier, Ch. 24.

Autonomic Control of the Heart & Endocrine Secretion of ANP and BNP

The heart is both a receptor of neural regulation and an active endocrine gland releasing natriuretic peptides to protect against volume overload.

SYMPATHETIC STIMULATION

Fight-or-Flight Acceleration

Neural Pathway: Postganglionic sympathetic fibers originate from T1-T4 cervical & upper thoracic ganglia (cardiac accelerator nerves).

Receptor Dynamics: Releases Norepinephrine (and adrenal Epinephrine) acting on $_x0008_eta_1$-adrenergic Gs-protein coupled receptors.

Four Cardinal Effects: • Positive Chronotropy: $\uparrow$ SA node firing rate (I_f current).

• Positive Inotropy: $\uparrow$ myocardial contractility (Ca2+ release via RyR2).

• Positive Dromotropy: $\uparrow$ AV nodal conduction speed.

• Positive Lusitropy: $\uparrow$ speed of relaxation (phospholamban phosphorylation).

Cardiac Reserve: Enables cardiac output to escalate 4-fold to 5-fold during acute physiological stress.

PARASYMPATHETIC BRAKE

Rest-and-Digest Deceleration

Neural Pathway: Preganglionic fibers travel via the Right & Left Vagus Nerves (Cranial Nerve X).

Receptor Dynamics: Releases Acetylcholine (ACh) onto M_2 muscarinic Gi-protein coupled receptors in SA and AV nodes.

Physiological Effects: • Negative Chronotropy: Activates $I_{K,ACh}$ channels causing hyperpolarization; reduces heart rate.

• Negative Dromotropy: Slows AV nodal conduction; prolonged high vagal tone can induce AV block.

• Minimal Ventricular Inotropy: Ventricles have scarce direct parasympathetic innervation.

Resting Vagal Tone: Prevents the heart from racing at its intrinsic SA node rate of 100 bpm, keeping healthy resting heart rate between 60 - 80 bpm.

HEART AS AN ENDOCRINE GLAND

ANP & BNP Natriuretic Peptides

Atrial Natriuretic Peptide (ANP): 28-amino-acid peptide synthesized and stored in granules within atrial myocytes. Secreted upon atrial mechanical stretch caused by hypervolemia.

B-Type Natriuretic Peptide (BNP): 32-amino-acid peptide produced primarily in ventricular myocytes in response to ventricular wall stress, pressure overload, and volume expansion.

Receptors & Second Messenger: Bind to NPR-A receptors containing guanylyl cyclase domains, boosting intracellular cyclic GMP (cGMP).

Physiological Counter-Regulation: • Renal: Increases GFR and inhibits Na+ reabsorption, driving natriuresis and diuresis.

• Vascular: Induces systemic arterial vasodilation, lowering BP.

• Endocrine: Potently suppresses Renin, Aldosterone, and Endothelin.

• Diagnostic: NT-proBNP is the universal biomarker for Heart Failure.

Coronary Perfusion Dynamics & Myocardial Energy Substrates

The heart is an omnivorous metabolic engine extracting 75% of arterial O2; Left ventricular myocardium receives perfusion almost exclusively during diastole.

CORONARY ARTERIAL ANATOMY

Blood Supply to the Pump

Arterial Origins: Arise from the left and right sinuses of Valsalva immediately above the aortic valve cusps.

Left Coronary System: Left Main breaks into Left Anterior Descending (LAD - 'widowmaker', supplies anterior wall & septum) and Circumflex (LCx, supplies lateral LV).

Right Coronary System: Right Coronary Artery (RCA) runs in atrioventricular groove, giving off Marginal branch and Posterior Descending Artery (PDA in 85% right-dominant hearts).

Nodal Perfusion: SA nodal artery arises from RCA (60%) or LCx (40%); AV nodal artery arises from RCA in 90% of individuals.

PHASIC CORONARY FLOW DYNAMICS

The Diastolic Perfusion Imperative

Systolic Compression: During ventricular systole, powerful intramyocardial pressure exceeds intracoronary pressure, squeezing intramural vessels shut (flow drops to near zero!).

Diastolic Window: 80% of Left Ventricular coronary blood flow occurs during DIASTOLE, when the myocardium relaxes and aortic diastolic recoil forces blood into coronary ostia.

Clinical Hazard of Tachycardia: Extreme heart rates (>140 bpm) shorten diastolic filling time far more than systolic time, severely compromising coronary perfusion and triggering myocardial ischemia.

Autoregulation: Local metabolites (Adenosine, NO, H+, pO2 drop) regulate coronary arteriolar resistance to match myocardial O2 demand.

METABOLIC SUBSTRATE PREFERENCE

ATP Production in Cardiac Mitochondria

Mitochondrial Density: Mitochondria occupy 30 - 40% of adult cardiomyocyte volume, reflecting absolute reliance on aerobic respiration.

Substrate Hierarchy at Rest: • Free Fatty Acids (FFA): 60 - 70% of ATP generated via $_x0008_eta$-oxidation.

• Glucose & Pyruvate: 20 - 25% of energy needs.

• Lactate & Ketone Bodies: 10% under resting conditions.

Metabolic Flexibility: During exercise, myocardium readily extracts and oxidizes circulating lactate produced by working skeletal muscles!

Extreme O2 Extraction: Myocardium extracts 70 - 80% of available arterial oxygen at rest (coronary sinus pO2 is only ~20 mmHg); thus, increased workload requires increasing blood flow.

Cardiovascular Metabolism: Opie, L. H., Heart Physiology: From Cell to Circulation, 4th Ed., Lippincott Williams & Wilkins.

Hemodynamic Reference Values & Diagnostic Cardiac Biomarkers

Quantitative clinical monitoring of pressure, flow, wall stress, and myocyte necrosis guides critical care management.

Parameter / Biomarker	Normal Reference Range	Physiological Significance	Clinical Diagnostic Role	Target Organ Affected
Cardiac Output (CO)	4.0 - 8.0 L/min	Total volume ejected by ventricle per minute (CO = HR x SV)	Identifies cardiogenic shock, low-output states, sepsis	All systemic organs
Mean Arterial Pressure (MAP)	70 - 105 mmHg	Average organ perfusion pressure over complete cycle	MAP <65 mmHg causes irreversible renal & cerebral ischemia	Brain, Kidneys, Viscera
Central Venous Pressure (CVP)	2 - 6 mmHg (3 - 8 cm H2O)	Right atrial filling pressure; assesses intravascular volume	Elevated in right ventricular failure and hypervolemia	Liver (congestive hepatopathy)
High-Sensitivity Troponin I / T	<14 ng/L (assay dependent)	Structural regulatory protein of cardiomyocyte myofibrils	Gold standard for Acute Myocardial Infarction (AMI)	Myocardium
NT-proBNP / BNP	<125 pg/mL (<300 pg/mL in acute dyspnea)	Cleaved neurohormone secreted during ventricular wall strain	Differentiates cardiac dyspnea from pulmonary disease	Heart, Kidneys, Vasculature
Ejection Fraction (LVEF)	55% - 70%	Fraction of end-diastolic volume ejected per beat (SV / EDV)	Classifies heart failure (HFrEF vs. HFpEF)	Whole-body circulation
Creatine Kinase-MB (CK-MB)	<5 ng/mL (or <3-5% of total CK)	Isoenzyme present in high concentration in myocardium	Detects re-infarction due to rapid 48-72h clearance window	Cardiac muscle

Cardiovascular Disease Cascades: From Ischemia to End-Organ Failure

Failure of the central pump triggers devastating multi-organ failure through forward hypoperfusion and backward venous congestion.

ISCHEMIC CASCADE & INFARCTION

Coronary Occlusion to Necrosis

Trigger: Rupture of an unstable atherosclerotic plaque with superimposed platelet thrombosis occludes a coronary artery branch.

Timeline of Injury: • Within 60 seconds: Myocyte ATP drops; systolic contractility ceases immediately.

• Within 10 minutes: Massive edema, loss of membrane potential, arrhythmia risk.

• Within 20 - 40 minutes: Irreversible coagulation necrosis begins from subendocardium to subepicardium (wavefront of ischemia).

Clinical Outcome: ST-elevation MI (STEMI), loss of functional contractile mass, ventricular arrhythmias (V-fib), or free wall rupture.

CONGESTIVE HEART FAILURE (CHF)

Neurohormonal Maladaptive Spiral

Initial Insult: Loss of myocyte contractility (HFrEF) or stiff, non-compliant filling (HFpEF) reduces forward cardiac output.

Vicious Compensatory Loop: Hypoperfusion triggers sympathetic hyperactivity and excessive RAAS activation:

• Chronic Norepinephrine induces myocyte apoptosis and fibrosis.

• Angiotensin II & Aldosterone promote toxic cardiac remodeling.

• Fluid retention elevates venous pressures.

Clinical Manifestation: Left-sided failure -> Pulmonary venous congestion, dyspnea, orthopnea, pulmonary edema. Right-sided failure -> Jugular venous distension, hepatosplenomegaly, peripheral edema.

CARDIORENAL & SHOCK CASCADES

Terminal Multi-Organ Dysfunction

Cardiorenal Syndrome (Type 1 & 2): Drop in cardiac output + elevated renal venous pressure (P_{rv}) compresses renal tubules, collapses GFR, and precipitates acute oliguric kidney injury.

Cardiogenic Shock: Sustained hypotension (SBP <90 mmHg, Cardiac Index <2.2 L/min/m²) despite adequate filling pressures.

Multi-Organ Failure (MODS): • Brain: Hypoxic ischemic encephalopathy, coma.

• Liver: Centrilobular necrosis ('nutmeg liver' & acute ischemic hepatitis).

• Gut: Mesenteric ischemia, translocation of endotoxins causing systemic inflammation.

Mortality: Cardiogenic shock carries 40 - 50% in-hospital mortality without rapid mechanical circulatory support (ECMO/Impella).

The 5 Fundamental Uses of the Heart Organ in the Human Body

Summary Blueprint: The heart integrates mechanical pumping, gas transport, fluid balance, metabolic cleansing, and endocrine signaling to preserve human life.

1. PRESSURE ENGINE

Continuous Pressure Gradient

Function: Maintains a constant hydrostatic pressure head (MAP ~93 mmHg) to drive continuous blood flow across high-resistance capillary networks.

Key Mechanics: Elastic arterial recoil (Windkessel effect) cushions high systolic pressure and preserves diastolic forward flow.

Organ Benefit: Guarantees capillary filtration across all vascular beds without stagnant stasis.

2. RESPIRATORY CONDUIT

O2 & CO2 Gas Transportation

Function: Pumps entire blood volume through alveolar membranes every minute, delivering 1,000 mL O2/min to peripheral tissues.

Key Mechanics: Maintains hemoglobin saturation at 98% and rapid transit of carbon dioxide (as bicarbonate and carbamino compounds) to lungs.

Organ Benefit: Prevents anaerobic lactic acidosis and cellular energy failure in all vital organs.

3. METABOLIC HIGHWAY

Nutrient & Waste Cleansing

Function: Connects intestinal nutrient absorption with cellular metabolism and organs of excretion.

Key Mechanics: Circulates glucose, fatty acids, amino acids, trace minerals, vitamins, and pharmacological compounds to target tissues.

Organ Benefit: Shunts nitrogenous wastes (urea, creatinine) to kidneys and hepatic toxins to liver.

4. ENDOCRINE SENSOR

Fluid Volume & Tone Governor

Function: Directly senses cardiovascular stretch and releases ANP and BNP hormones into circulation.

Key Mechanics: Activates particulate guanylyl cyclase, elevating cGMP to relax vascular smooth muscle and accelerate renal salt excretion.

Organ Benefit: Serves as the internal volume protector preventing acute pulmonary edema.

5. THERMAL & IMMUNE LINE

Body Defense & Heat Buffer

Function: Distributes core thermal energy to peripheral cutaneous capillary beds for evaporative and radiative heat loss.

Key Mechanics: Constricts in freezing environments to protect core cardiac and cerebral perfusion.

Organ Benefit: Rapidly transports neutrophils, lymphocytes, platelets, and immunoglobulins to injury sites within seconds.